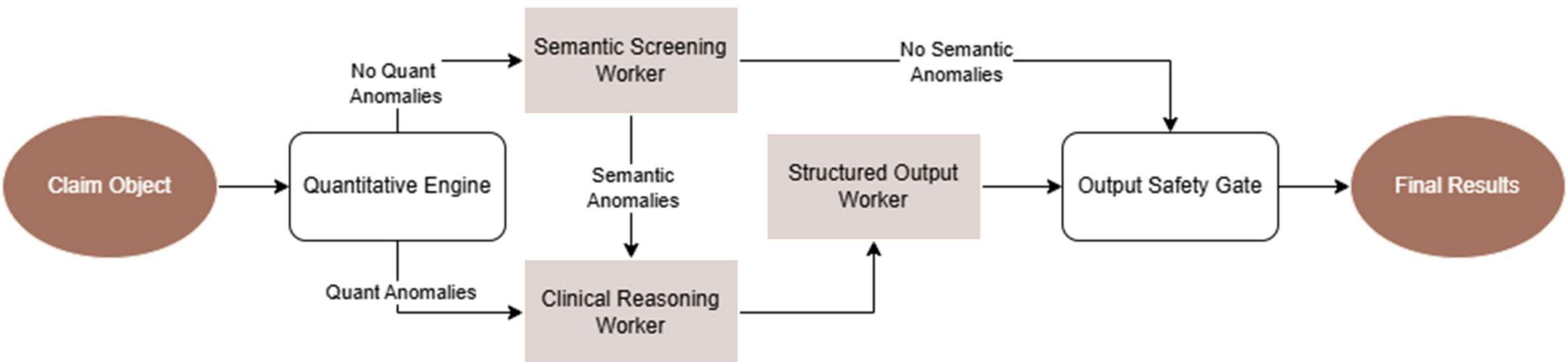
Saumya Bajaj

AGENTIC AI & PREDICTIVE ANOMALY DETECTION IN HEALTHCARE OPERATIONS

THE CORE GAPS IN MODERN TPO SYSTEMS

Legacy Billing Checkers	Unconstrained AI Agents
Blind to semantic intent Vulnerable to “silent leakage” (modifier abuse, context mismatch)	Excessively high compute latency Extreme token costs Prone to hallucination Fragmentation of safety boundaries

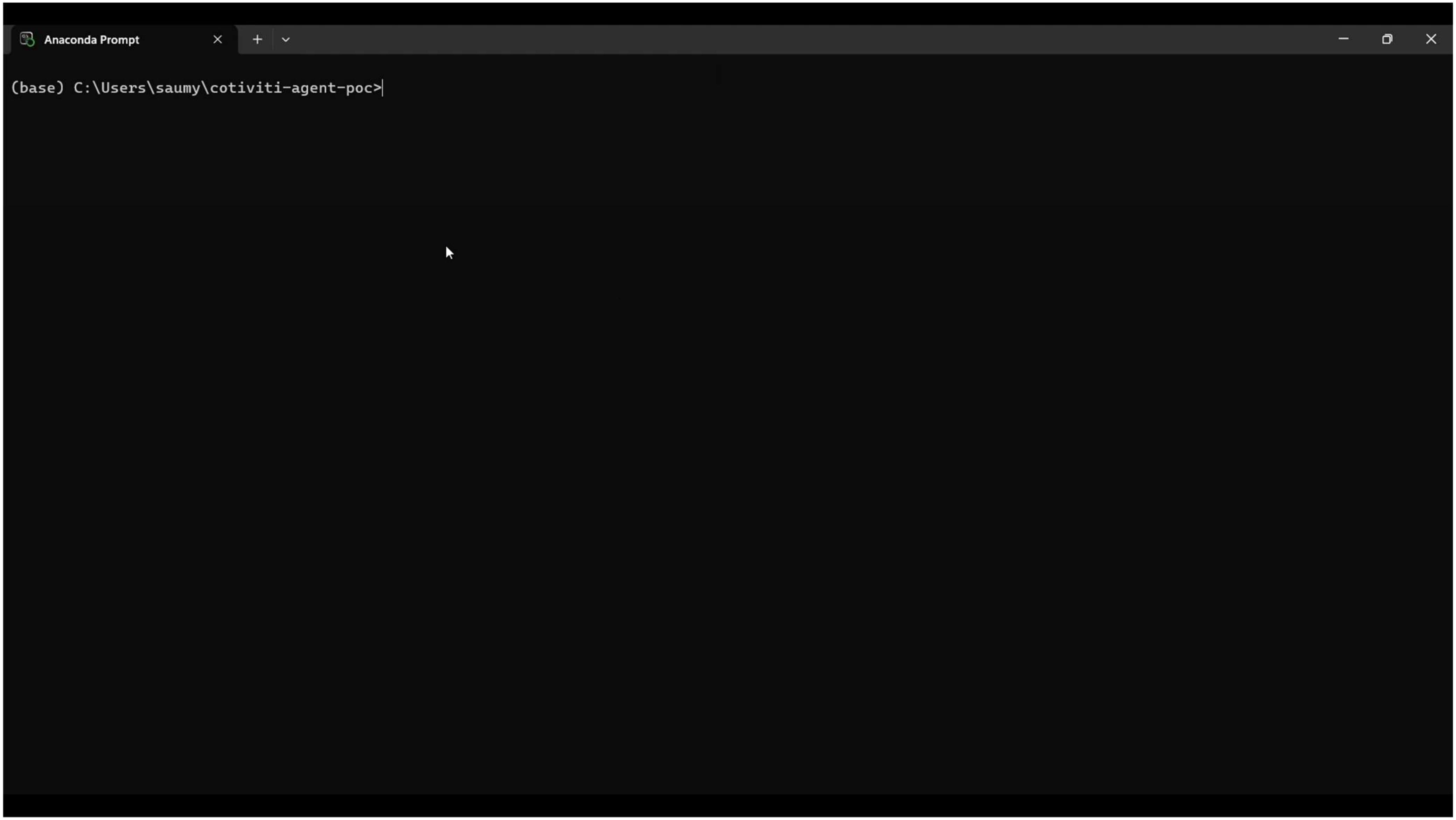
SOLUTION SYSTEM ARCHITECTURE



LIVE RUN

Processing a 10-claim structural database profile.

Core Challenge: Isolate unit ceilings, extreme statistical outliers, and complex contradictions simultaneously


Anaconda Prompt
(base) C:\Users\saumy\cotiviti-agent-poc>

ENTERPRISE STRATEGIC POSITIONING

Production Scalability:

Automated screening expands the auditable surface market from a 1% manual slice to 100%

Pattern Anchoring:

Transforms heavy semantic reasoning into efficient structural text-matching

R isk Mitigation

Strict Pydantic configurations ensure enterprise type-safety & eliminate model hallucinations

ROAD TO PRODUCTION

Phase 1 – Containerization & Dependency Freeze

Phase 2 – Event-Driven Eventual Consistency

Phase 3 – Automated Evaluation & Prompt CI/CD

Phase 4 –Telemetry & Cost-Guardrail Instrumentation

Ahmad, M., et al. (2026). Safety, hallucination, and failure modes in agentic health AI: A state-of-the-art review. IEEE/Preprints.

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SAS Institute. (2025). Agentic AI is the fix for communication breakdowns in health care payment integrity. SAS Voices.

Saxena, A., & Forbes Technology Council. (2026). FWA detection is shifting left: Agentic AI's role in program integrity. Forbes.

Velasquez, J., et al. (2026). Benchmarking large language model-based agent systems for clinical decision tasks. PubMed Central (PMC13031301).